四川大学期末考试试题（闭卷）

（2022——2023 学年第 1 学期）　A 卷

课程号：304032030　课序号：　课程名称：计算机网络（双语）　成绩：
任课教师：吕光宏 朱敏 张靖宇 陈黎 杨朝斌 傅静涛　学生人数：　印题份数：
适用专业年级：2020　学号：　姓名：　**（所有的答案写到答题纸上!!!!!!!!）**

考 生 承 诺
我已认真阅读并知晓《四川大学考场规则》和《四川大学本科学生考试违纪作弊处分规定（修订）》，郑重承诺： 1、已按要求将考试禁止携带的文具用品或与考试有关的物品放置在指定地点； 2、不带手机进入考场； 3、考试期间遵守以上两项规定，若有违规行为，同意按照有关条款接受处理。 **考生签名：**

I、**Acronyms** (5 points, 1 point for each acronym)

Consider the following acronyms: AIMD, BGP, CIDR, DHCP, DNS, FIN, HTTP, ICANN, MSS, MTU, NAT, RFC, CRC, RIP, RST, SYN, TCP, TLD, UDP.

Match the acronyms to the descriptions below, using each acronym exactly once. Supply exactly one answer for each item below. In some cases, an acronym might fit more than one question, but there is only one way to answer these so that every acronym is used exactly once.

1） An error-detecting code allows detection of corrupted data within the entire Ethernet frame. CRC

2） The name of the message used to initiate a TCP connection. SYN

3） The documents used by the IETF to describe protocol standards. RFC

4） A widely used routing protocol that does not necessarily compute lowest-cost paths. BGP.

5） A network management protocol used on Internet Protocol (IP) networks, which dynamically assigns an IP address and other network configuration parameters to each device on the network. DHCP

II、**Explanation of terms** (10 points, 2 points for each term)

1） Cumulative ACK　2）Best effort service　3）CSMA/CD　4）Socket　5）CIDR

III、**Single Choice** (10 points, 1 point for each question)

1） In pipelining protocols, the selective repeat approach requires (　).

A、receiver acks individual packets

B、sender maintains timer for cumulative unacked packets

C、receiver only sends cumulative acks

D、both B and C responses are correct

2） Which application's NOT using TCP? (　　)

A、SMTP　　　　B、HTTP　　　　C、DNS　　　　D、All of them

3） What are the advantages of Distance Vector Routing over Link State Routing? (　　)

A、Changes faster, especially when links go down.

B、Requires less computation in each router.

C、The computed routers are of better quality.

D、Requires more information to be exchanged by the routers.

4） Suppose host A want to create a TCP connection with host B by sending a TCP segment which SYN=1 and seq=11220. Suppose that host B receive this request. The correct TCP segment which is sent back to host A should be (　　).

A、SYN=0，　ACK=0，　seq=11221，　ack=11221

B、SYN=1，　ACK=1，　seq=11220，　ack=11220

C、SYN=1，　ACK=1，　seq=11221，　ack=11221

D、SYN=0,　ACK=0，　seq=11220，　ack=11220

5） Consider the process which sending and receiving email between user1 and user2, as demonstrated in Figure 1. The application layer protocol for ①②③ respectively is (　　).

A、SMTP、SMTP、SMTP　　　　B、POP3、SMTP、POP3

C、POP3、SMTP、SMTP　　　　D、SMTP、SMTP、POP3

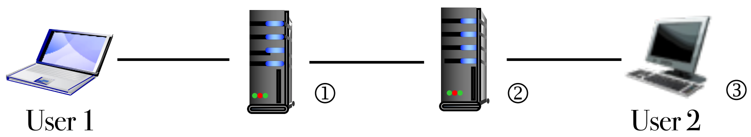


Figure 1

A routing table contains four routing entries with the same forwarding interface, whose destination network addresses are 35.230.32.0/21, 35.230.40.1/21, 35.230.48.0/21, and 35.23.56.0/21. What is the destination network address of these four routes after they are aggregated? (　　)

A、35.230.0.0./19　　　　B、35.230.0.0/20

C、35.230.32.0/19　　　　D、35.230.32.0/20

6） The header field based on which the UDP protocol implements demultiplexing is (　　).

A、source port number　　B、destination port number　　C、length　　D、checksum

7） Which of the following statements about the FTP protocol is false? (　　).

A、The data connection is closed after each data transfer

B、The control connection remains open for the duration of the session

C、The server establishes a data connection with TCP port 20 of the client

D、The client establishes a control connection with TCP port 21 of the server

8） The HTTP request packet sent by a browser is shown in Figure 2. In the following statement, (　　) is wrong.

A、 The browser requests to browse index.html

B、 Index.html is available at www.test.edu.cn

C、 The browser requests a persistent connection

D、 This browser has been visited www.test.edu.cn

```
GET /index.html HTTP/1.1
Host: www.test.edu.cn
Connection: Close
Cookie: 123456
```

Figure 2

9） Table 1 shows the routing table of a router. If a router receives an IP packet whose destination address is 169.96.40.5, the interface that forwards the IP packet is (   ).

A、S1   B、S2   C、S3   D、S4

Table 1

Destination Network	Next Hop	Interface
169.96.40.0/23	176.1.1.1	S1
169.96.40.0/25	176.2.2.2	S2
169.96.40.0/27	176.3.3.3	S3
0.0.0.0/0	176.4.4.4	S4

10） Which of the following is a random-access protocol? (   )

A、CSMA/CD   B、Polling   C、TDMA   D、ARP

IV、Decide true or false, give a brief explanation for each one. (15 points, 3 points for each question)

1） Assume the propagation delay in a broadcast network is 3μs and the frame transmission time is 5μs. The collision can be detected no matter where it occurs.   (F)

解答：F。The sender needs to detect the collision before the last bit of the frame is sent out. If the collision occurs near the destination, it takes 2 × 3 = 6 μs for the collision news to reach the sender. The sender has already sent out the whole frame; it is not listening for a collision anymore.

2） A IP address like 255.255.255.12 can be a mask in CIDR.   (F)

解答：F。We first write each potential mask in binary notation and then check if it has a contiguous number of 1s from the left followed by 0s. 255.255.255.12 is 11111111 11111111 11111111 00001100. So it can not be a mask.

3） Rather than counting packets, TCP sequence numbers count bytes in the byte stream . (T)

4） One role of a NIC is to accept a datagram from the next higher layer and encapsulate the datagram into a frame, along with frame header and trailer bit fields, such as rdt and CRC.   (T)

5） In packet-switched networks, the resources (buffers, link transmission rate) needed along a path to provide for communication between the end systems are reserved for the duration of the communication session between the end systems. (F) no reservation

V、Integrated Questions （60 points)

1） TCP provides a reliable data transfer service that is implemented on top of an unreliable IP end-to-end network layer. Please try to describe the TCP's reliable data transfer mechanism and give out the FSM for the sender and for the receiver under channel 2.1.   (8 Points)

给分标准：描述 RDT 基本原理（2 分），给出 FSM 的基本形式（1 分）/给出发送方和接收方各自的基本图形描述（2 分）；指出 ACK/NAK 丢失或出错后的处理（2 分）；指出帧序号以及与 ACK/NAK 的配合（2 分）。

2） Compare virtual circuit and datagram networks. (6 Points)

A、 Fill the blank in the following table 2 using yes or no.

B、 Compare the advantages and disadvantages about them.

Table 2

Criterion	Datagram Networks	Virtual Circuit Networks
A path has been established before the packets are sent.	×	√
Error Control and Flow Control	×	√
In-order Packets Transfer	×	√
Packet Loss-tolerant	√	×
The destination ID is needed to delivery packets	√	×

给分标准：填写 YES/NO，Y/S，或用√/×表示均可，上表共 5 分，每个空 0.5；第二问根据上述表格进行优缺点比较，得 1 分。

3） Give some protocols as examples, describe the differences between random access protocols and taking-turn protocols. (5 Points)

给分标准：对两种协议给出例子，得 1 分；分别对两种协议进行原理解释，得 2 分；对两种协议进行比较，得 2 分。

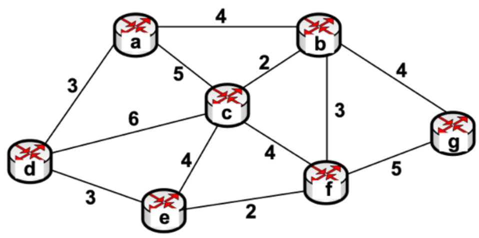
4） Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why? (6 points)

Web caching can bring the desired content “closer” to the user, possibly to the same LAN to which the user’s host is connected. Web caching can reduce the delay for all objects, even objects that are not cached, since caching reduces the traffic on links.

第一问能够解释清楚什么是缓存，给 3 分。第二问答案为 all objects，且大致说清楚正确理由给 3 分；答案为 some of the objects，但能够说出合适的理由，酌情给 1-2 分。

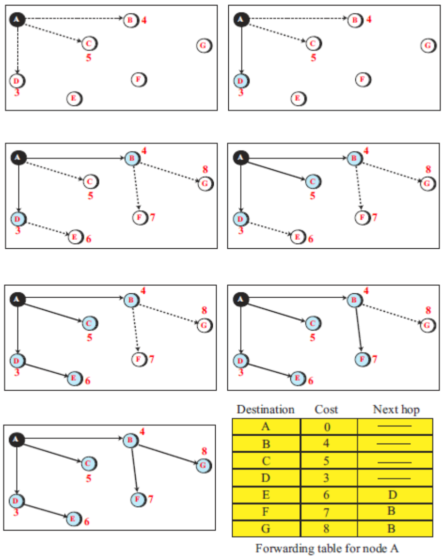
5） Consider the network topology shown in the below figure 3.

A、 Use Dijkstra’s algorithm to find the shortest path tree and the forwarding table for node a. (8 Points)



解答：见右图，也可以使用表格的形式，可参考 Kurose，英文版教材（E4），P399，Figure4.3。给分标准：最短路径或最短路径树 6 分，节点 a 的转发表 2 分。

Figure 3



Destination	Cost	Next hop
A	0	——
B	4	——
C	5	——
D	3	——
E	6	D
F	7	B
G	8	B

Forwarding table for node A

B、 Based on the above calculations, if we want to find the shortest path for a single node when the number of nodes is n, please calculate the number of searches. (4 Points)

解答：The number of searches in each iteration of Dijkstra's algorithm is different. In the first iteration, we need n number of searches, in the second iteration, we need (n - 1), and finally in the last iteration, we need only one. In other words, the total number of searches for each node to find its own shortest-path tree is

Number of searches = n + n − 1 + n − 2 + n -3 +···+3 + 2 + 1 = n (n + 1) / 2

The series can be calculated if it is written twice: once in order and once in the reverse order. We then have n items, each of value (n + 1), which results in n(n+ 1). However, we need to divide the result by 2. In computer science, this complexity is written as $O(n^2)$ and is referred to as Big-O notation.

6） According to the Figure 4, describe the congestion control policy in TCP. (9 Points)

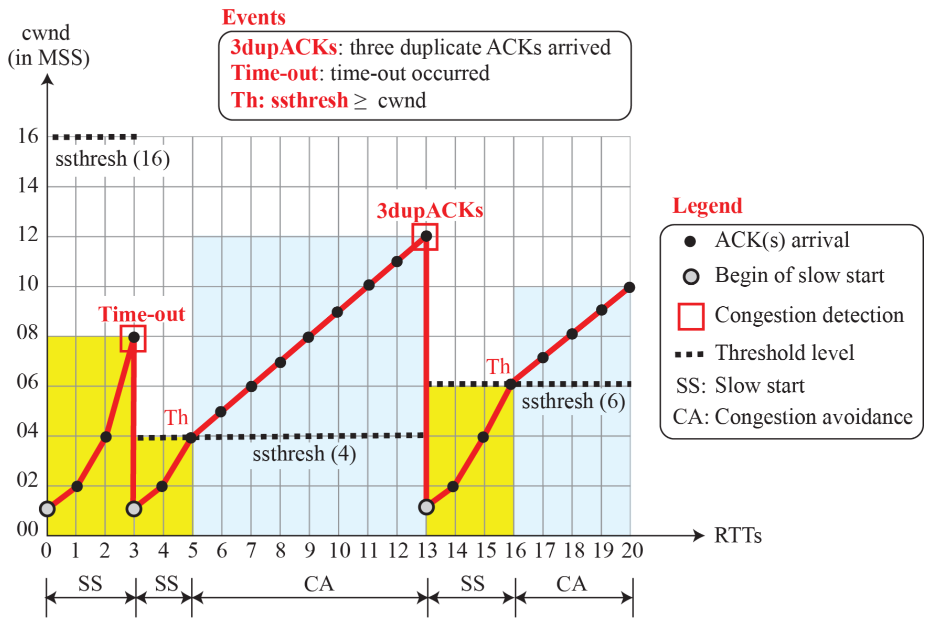


Figure 4

对照拥塞控制的基本原理，要覆盖图中红色文字的要点，以及右侧图例中的原理（慢启动、AIMD、Threshold 的变化、拥塞避免等）

7） According to the figure 5, describe the process of the communication through an internet, including the protocol suite, logical connection, switch and etc. (6 Points)

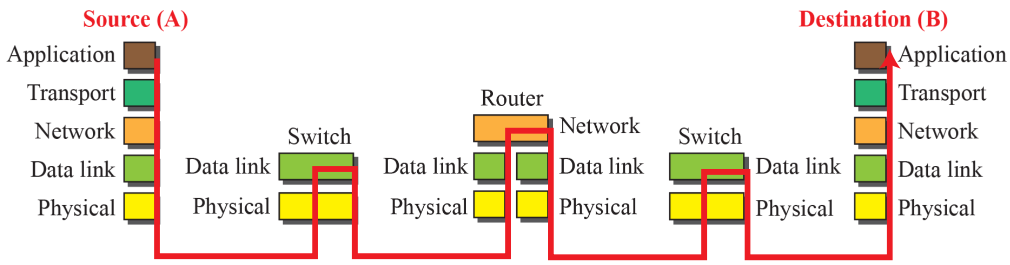


Figure 5

8） Suppose two nodes, A and B, are attached to opposite ends of a 900m cable, and that they each have one frame of 1000bits (including all headers and preambles) to send to each other. Both nodes attempt to transmit at time t=0. Suppose there are four repeaters between A and B, each inserting a 20-bit delay. Assume the transmission rate is 10Mbps, and CSMA/CD with backoff intervals of multiples of 512 bits is used. After the first collision, A draws K=0 and B draws K=1 in the exponential backoff protocol. Ignore the jam signal and the 96-bit time delay. (8 Points)

A、 What is the one-way propagation delay (including repeater delays) between A and B in seconds. Assume that the signal propagation speed is $2 \cdot 10^8$ m/sec.

B、 At what time (in seconds) is A's packet completely delivered at B?

解答:

A $= \dfrac{900m}{2\times10^8 m/\sec} + 4\times\dfrac{20bits}{10^7 bps} = 12.5\mu s$ （4 分）

B At time t = 0, both A and B transmit. A detects collision at timet = 12.5μs. A picks K = 0 and (ignoring jam signal) waits for the channel to be idle. The channel becomes idle at time t = 25μs. This is the time last bit from B arrives at A. A retransmits at t = 25μs (ignoring 96-bit time delay). A finishes transmission at time t = 125μs. Last bit from A arrives at B at time 137.5μs. （4 分）